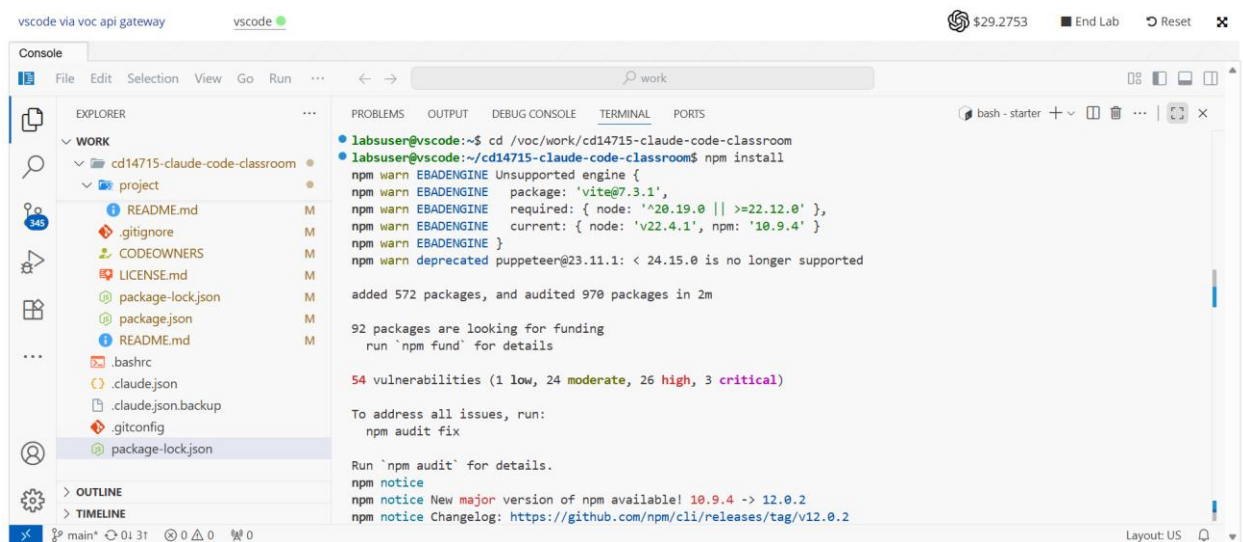


NOTE: Paste (Ctrl + v) will not work inside the Terminal, Use Shift + Insert for pasting commands or anything inside the Terminal.

1. Go to your Vocareum lab and click Start Cloud Resource, then Open Cloud Console to launch the VS Code workspace.
2. Open a terminal inside it.
3. If you see a folder named cd14715-claude-code-classroom, it's already cloned, execute below commands. Otherwise, clone it & then run below commands.
 - `cd /voc/work/cd14715-claude-code-classroom`
 - `npm install`

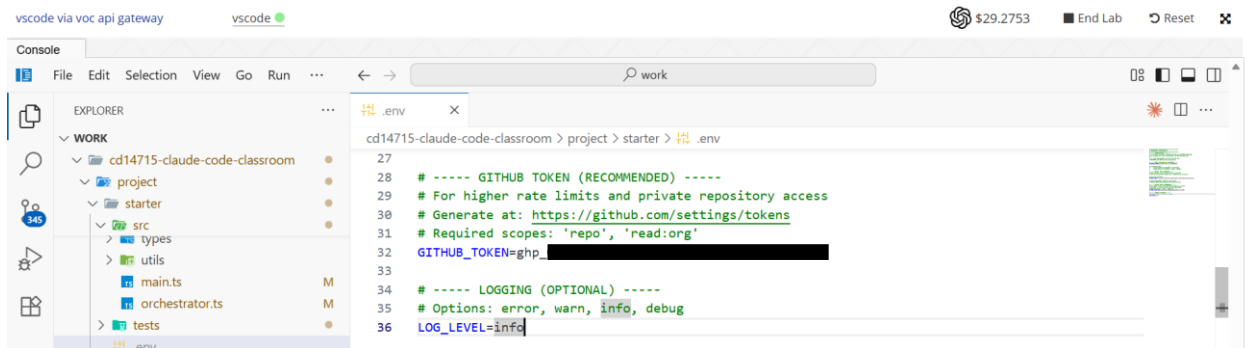


4. Configure your environment
 - `cd /voc/work/cd14715-claude-code-classroom/project/starter`
 - `cp .env.example .env`
5. Do not add ANTHROPIC_API_KEY or ANTHROPIC_BASE_URL — Vocareum already has those set in your workspace environment. You can confirm with:
 - `echo $ANTHROPIC_API_KEY | head -c 10`
 - `echo $ANTHROPIC_BASE_URL`

```
labsuser@vscode:~/cd14715-claude-code-classroom$ cd project/starter
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$ echo $ANTHROPIC_API_KEY | head -c 10
voc-143214labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$ echo $ANTHROPIC_BASE_URL
https://claude.vocareum.com
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$ cp .env.example .env
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$ echo "ANTHROPIC_API_KEY=$ANTHROPIC_API_KEY"
ANTHROPIC_API_KEY=voc-143214labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$ echo "ANTHROPIC_BASE_URL=$ANTHROPIC_BASE_URL"
ANTHROPIC_BASE_URL=https://claude.vocareum.com
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$
```

6. Get a GitHub token (recommended, avoids rate-limit failures on the real PR fetch):

- Go to <https://github.com/settings/tokens>
- Generate a new token (classic) with scopes repo and read:org
- Paste it into .env as GITHUB_TOKEN=ghp_...



7. Complete the code in starter's TODO stubs. In src, tests & .claude folder.

8. Verify the build, from /voc/work/cd14715-claude-code-classroom/project/starter:

This should complete with no errors.

- npm run build

```
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$ npm run build
```

```
> claude-multiagent-code-review-starter@1.0.0 build  
> tsc  
  
npm notice  
npm notice New major version of npm available! 10.9.4 -> 12.0.2  
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2  
npm notice To update run: npm install -g npm@12.0.2  
npm notice
```

```
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$
```

9. Run the tests. This runs tests/orchestrator.test.ts, which mocks the Claude Agent SDK so it doesn't need real API access — it should pass immediately and confirms the orchestrator wires up MCP servers, all 3 subagents, and schema validation correctly.

- npm test

```

    ✓ Configuration (2)
      ✓ should initialize with default options 2ms
      ✓ should accept custom rate limit configuration 1ms
    ✓ reviewPullRequest (5)
      ✓ should fetch PR files from GitHub MCP 6ms
      ✓ should spawn all 3 subagents in parallel 2ms
      ✓ should aggregate results into ReviewReport 2ms
      ✓ should validate output with Zod schema 2ms
      ✓ throws a clear error when the SDK never produces a structured_output 3091ms
    ↓ Integration (1)
      ↓ should review a real small PR

Test Files  1 passed (1)
Tests       7 passed | 1 skipped (8)
Start at    15:46:34
Duration    4.23s (transform 194ms, setup 0ms, import 884ms, tests 3.11s, environment 0ms)

npm notice
npm notice New major version of npm available! 10.9.4 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0.2
npm notice
labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$

```

10. Smoke-test against a real PR

- npm run dev -- octocat Hello-World 1

```

○ labsuser@vscode:~/cd14715-claude-code-classroom/project/starter$ npm run dev -- octocat Hello-World 1

> claude-multiagent-code-review-starter@1.0.0 dev
> tsx src/main.ts octocat Hello-World 1

🔑 Using Anthropic API authentication
2026-09-02 15:48:05 [info]: Starting review of octocat/Hello-World PR #1...
2026-09-02 15:48:05 [info]: Starting code review
{
  "service": "code-review-system",
  "owner": "octocat",
  "repo": "Hello-World",
  "prNumber": 1
}

2026-09-02 15:50:46 [info]: Code review completed
{
  "service": "code-review-system",
  "owner": "octocat",
  "repo": "Hello-World",
  "prNumber": 1,
  "score": 35,
  "duration": 161148,
  "status": "success"
}
2026-09-02 15:50:46 [info]: Review complete. Reports saved:
2026-09-02 15:50:46 [info]: JSON:      reports/octocat_Hello-World_1.json
2026-09-02 15:50:46 [info]: Markdown: reports/octocat_Hello-World_1.md
2026-09-02 15:50:46 [info]: HTML:      reports/octocat_Hello-World_1.html
2026-09-02 15:50:46 [info]: Overall score: 35/100

```

11. Generate the required submission reports

- npm run dev -- airaamane simple-todo-app 1
- npm run dev -- airaamane simple-todo-app 2
- npm run dev -- airaamane simple-todo-app 3

```
> claude-multiagent-code-review-starter@1.0.0 dev
> tsx src/main.ts airaamane simple-todo-app 1
```

```
🔒 Using Anthropic API authentication
2026-09-02 15:56:59 [info]: Starting review of airaamane/simple-todo-app PR #1...
2026-09-02 15:56:59 [info]: Starting code review
{
  "service": "code-review-system",
  "owner": "airaamane",
  "repo": "simple-todo-app",
  "prNumber": 1
}
```

```
> claude-multiagent-code-review-starter@1.0.0 dev
> tsx src/main.ts airaamane simple-todo-app 2
```

```
🔒 Using Anthropic API authentication
2026-09-02 16:05:01 [info]: Starting review of airaamane/simple-todo-app PR #2...
2026-09-02 16:05:01 [info]: Starting code review
{
  "service": "code-review-system",
  "owner": "airaamane",
  "repo": "simple-todo-app",
  "prNumber": 2
}
```

```
🔒 Using Anthropic API authentication
2026-09-02 16:13:03 [info]: Starting review of airaamane/simple-todo-app PR #3...
2026-09-02 16:13:03 [info]: Starting code review
{
  "service": "code-review-system",
  "owner": "airaamane",
  "repo": "simple-todo-app",
  "prNumber": 3
}
2026-09-02 16:13:35 [info]: Code review completed
{
  "service": "code-review-system",
  "owner": "airaamane",
  "repo": "simple-todo-app",
  "prNumber": 3,
  "score": 0,
  "duration": 32177,
  "status": "success"
}
```

If you are getting score as zero, try below:

- `npm run dev -- shinshin86 todo-opfs-sqlite 1`
- `npm run dev -- lucaong minisearch 295`
- `npm run dev -- lucaong minisearch 305`

12. Either you upload the zip file or use GitHub URL.

For Reference:

<https://github.com/NeeharikaRN/Neeha-Bounded-Autonomy-and-Guardrails-with-Claude-and-Claude-Code.git>